Safety-First Maternal Risk Screening: Documented Clinical Thresholds as a Baseline, Label-Transfer Calibration, and Urdu Risk Communication as a Deliverable

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Abstract*—Maternal risk classification on the UCI Maternal Health Risk dataset has converged on a single research shape: preprocess, train a tree ensemble, report accuracy in the low-to-mid eighties, and conclude that machine learning is promising for maternal healthcare. We argue that this shape leaves three gaps unaddressed. It optimises a metric that cannot distinguish the two error types that matter clinically; it treats risk labels produced by an unpublished protocol in one country as ground truth for deployment in another; and it stops at a number, when the artefact a community health worker needs is a sentence she can say to a family. We restructure the task around those gaps and contribute (i) a safety-first evaluation protocol built on a separately reported *critical-miss* rate, an asymmetric cost matrix, and referral thresholds tuned under an operational budget nested inside the cross-validation loop; (ii) a label-transfer calibration study encoding eleven sourced obstetric criteria as a transparent rule baseline; and (iii) hand-authored Urdu risk communication whose safety properties are enforced by continuous-integration lint rules rather than reviewer diligence. Our headline result is negative and, we argue, the most useful finding available here: no learned model beat the eleven documented thresholds on high-risk recall, even after equalising referral load (0.918 versus 0.847 for the best learned model). The learned models earn their place differently, cutting the catastrophic high-to-low error from 3.5% to 1.2% or below and improving balanced accuracy from 0.642 to 0.700, which supports a hybrid rules-plus-model design that the accuracy-reporting framing never puts on the table. We further find that the sampling condition of the dataset's *undocumented* blood-glucose column moves the clinical baseline more than any modelling choice available to us: reading it as fasting rather than post-load glucose shifts guideline–label agreement from 64.0% to 31.4%.

***Index Terms*—Clinical decision support, cost-sensitive learning, explainable artificial intelligence, external validity, health informatics, low-resource languages, maternal health, model calibration.**

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Provenance notice. Every quantitative result reported in this manuscript was computed on a deterministic *synthetic* stand-in for the UCI Maternal Health Risk dataset, not on the real file. The results demonstrate that the pipeline, the metrics, and the analyses behave as specified; they are *not* findings about maternal health. Section IV-B explains the construction and the reasoning. Claims whose direction could plausibly reverse on real data are flagged in place.

Not a medical device. The described system is a research prototype. It is not registered or certified with any regulatory authority, has not been validated on any patient population, does not diagnose any condition, and must not be used to make clinical decisions.

Code, data card, and the full artefact set are available at https://github.com/laiba-mazhar/maternal-health-risk.

I. Introduction

PAKISTAN'S maternal mortality ratio remains among the higher figures in South Asia [1], and the majority of maternal deaths worldwide are caused by conditions detectable in advance: hypertensive disorders, haemorrhage, sepsis, and complications of pre-existing disease [2]. In much of rural Pakistan the first clinical contact is not an obstetrician but a Lady Health Worker (LHW) [3], [4], who can take a blood pressure, a temperature, a pulse, and a glucose strip, and who must then make a single decision: refer, or not.

That decision is the intervention. Anything a screening tool does that does not improve it — including its accuracy figure — is decoration. This framing drives every choice in this paper and is where we depart from existing work.

A. Three Properties of a Useful Screening Tool

1) It must weight errors the way the clinic does: Roughly 37% of this cohort is labelled low risk, so a classifier predicting "low risk" for every mother attains 37% accuracy while missing 100% of high-risk cases. This is not a straw man: it is the direction any unweighted loss pushes a model on imbalanced data. Worse, accuracy treats *predicting mid risk for a high-risk mother* — which still escalates her — as identical to *predicting low risk*, which sends her home. Those errors differ in kind, not degree.

2) It must not assume its labels transfer: The dataset was collected through a community health-monitoring initiative in Bangladesh [5], and its labelling protocol was never published: we do not know which guideline, if any, the labellers applied, nor how disagreements were resolved. Treating those labels as ground truth for a Pakistani deployment embeds an untested transfer assumption at the root of the model — the failure mode systematic reviews of clinical prediction models find repeatedly [6], [7].

3) Its output must be deliverable at the point of care: A probability, or a risk band in clinical English, is not something an LHW can say to a family in rural Punjab. The last mile — a sentence that is understood, believed, and acted upon — is where a screening programme succeeds or fails [8], and it is almost entirely absent from the machine-learning literature on this dataset.

B. Contributions

1) A **safety-first evaluation protocol** (Section V-B) whose primary quantities are high-risk recall, a separately tracked critical-miss rate, expected cost under an asymmetric matrix, and referral load; in which the decision rule is separated from the classifier and tuned under an explicit referral budget *inside* the cross-validation loop; and in which model selection is on recall rather than accuracy.
2) A **label-transfer calibration study** (Section VI-B) encoding eleven sourced obstetric criteria as a transparent rule baseline, measuring agreement, directionality of disagreement, and per-criterion discriminative contribution.
3) A **matched-referral comparison** (Section VI-C) that equalises operational cost before comparing a threshold-tuned model against a fixed clinical rule — the only form of that comparison we consider honest.
4) **Urdu risk communication as a first-class, testable deliverable** (Section V-F), hand-authored rather than machine-translated, with three safety commitments enforced as executable lint rules.
5) A **reproducible artefact** in which every reported number is regenerated by a single command and stamped with its data provenance.
6) A **negative headline result** (Section VI-C) and an analysis of what the learned models do buy, which points to a hybrid design.

II. Related Work

A. Machine Learning on the UCI Maternal Health Dataset

Since the dataset's release [9], [5], a substantial body of work has applied the standard tabular toolkit to it: decision trees, random forests, gradient boosting, occasionally a small neural network, usually with SMOTE [10] or class weighting for the imbalance, typically reporting accuracy and macro F1 in the low-to-mid eighties. We are not aware of prior work on this dataset that reports a class-specific safety metric, separates the decision threshold from the classifier, or compares against a guideline-derived rule baseline.

B. Obstetric Risk Scoring

Clinical obstetric risk scoring predates machine learning and is built from documented thresholds rather than fitted parameters. That literature supplies the criteria we encode in Section V-A; what it does not supply is an account of what a learned model adds over it, which is the question Section VI-C asks directly.

C. Evaluation Methodology for Clinical Prediction Models

The methodological literature has been explicit for years that discrimination alone is insufficient, that calibration is where deployed models fail [7], [11], and that the decision-analytic question — what does acting on this model cost and gain — requires net-benefit reasoning rather than a threshold-free summary [12]. Our referral-budget constraint and matched-referral comparison are a coarse, discrete instance of that reasoning: rather than integrating net benefit across thresholds, we fix the operational cost and compare recall there. TRIPOD [13] asks for the provenance and transfer analysis that Section VI-B provides. Cost-sensitive learning provides the formal basis for the asymmetric matrix we adopt [14].

D. Explainability in Clinical Machine Learning

We use SHAP [15] in its exact forms: TreeSHAP for the ensembles [16] and the closed-form linear explainer for the logistic model. Our contribution here is epistemic rather than methodological: we require the attribution method to be named in every output, so a silent fallback to an approximate method cannot be presented as SHAP. Rudin's argument for interpretable models in high-stakes settings [17] is relevant to our outcome, since the model selected on safety grounds turned out to be the interpretable one.

E. Health Communication and Low-Resource Languages

Work on health communication establishes that comprehension, trust, and actionability determine whether advice is followed. The machine-learning literature on this dataset does not engage with it. Machine translation of clinical instructions is not a safe substitute, as a mistranslated instruction is a safety incident rather than a quality defect.

F. Data Documentation and Proxy Labels

Datasheets [18] presume that provenance is recoverable. Where it is not — as with the glucose sampling condition here — the practical recipe is to state the assumption and measure what it costs, which is what Section VI-B does. Obermeyer *et al.* [19] showed how a proxy label can encode a systematic inequity that accuracy cannot see; the analogue here is milder but structurally identical.

III. Gap Analysis and Positioning

Table I consolidates the gaps left by prior work and maps each to the section addressing it. Our aim is explicitly *not* to improve on published accuracy figures — Section VI-A shows our models land in the same band. It is to argue that the figure was never the interesting quantity.

IV. Data

A. The Real Dataset, and One Load-Bearing Ambiguity

The UCI Maternal Health Risk Data Set contains approximately $1{,}014$ records with six features — age, systolic and diastolic blood pressure, blood glucose, body temperature, and resting heart rate — and an ordinal label in $\{\text{low}, \text{mid}, \text{high}\}$ risk, distributed roughly $406/336/272$. The imbalance runs in the worst direction: the class we least want to miss is the smallest.

Three properties shape everything downstream.

TABLE I
GAPS IN EXISTING WORK AND WHERE THIS PAPER ADDRESSES THEM

ID	Gap left by existing work	Addressed in
G1	Symmetric metrics cannot express that high→low is categorically worse than high→mid	Sec. V-B, VI-A
G2	Dataset labels assumed to transfer; provenance and units treated as settled	Sec. V-A, VI-B
G3	No clinical floor; no account of what learning adds over documented rules	Sec. V-A, VI-C
G4	Thresholds implicit (`argmax`) or tuned with leakage; referral load an accident	Sec. V-C, V-D
G5	"We applied SHAP" conceals method substitution and silent degenerate output	Sec. V-E, VI-D
G6	Localisation absent or machine-translated; message quality treated as a review activity	Sec. V-F, VI-F
G7	Documentation frameworks presume recoverable provenance	Sec. IV-A, VI-B

TABLE II
WHO THRESHOLDS FOR HYPERGLYCAEMIA IN PREGNANCY

Reading	Gestational diabetes	Diabetes in pregnancy
Fasting plasma glucose	5.1–6.9 mmol/L	≥ 7.0 mmol/L
2-hour 75 g OGTT	8.5–11.0 mmol/L	≥ 11.1 mmol/L

1) Units are undocumented, and one is load-bearing: Body temperature is Fahrenheit (reading it as Celsius would be catastrophic) and glucose is mmol/L, but the *sampling condition* of the glucose column is unrecorded. WHO thresholds for hyperglycaemia in pregnancy differ sharply by sampling condition [20], as shown in Table II.

The observed range is roughly 6–19 mmol/L with a median near 8. Read as fasting values, over 80% of a community *screening* cohort would be diabetic, which is not credible; read as post-load values the distribution is clinically ordinary. We therefore assume the OGTT reading, state the assumption explicitly, and report the sensitivity of every guideline result to it (Section VI-B) — following the spirit of [18] for the case where documentation is already unrecoverable.

2) Some values are physiologically impossible: Heart rates of 7 bpm are dropped digits, not bradycardic patients; ages up to 70 are not maternal ages. We repair these by class-conditional median imputation rather than dropping the row, preserving the five other usable vitals on an affected record.

3) Two clinical criteria cannot fire: Diastolic pressure never reaches the severe-hypertension threshold of 110 mmHg, and heart rate never exceeds 90 bpm, so the tachycardia criterion (> 100) is unreachable. We retain both and report them as *silent*: a criterion whose cut-off the data cannot reach is a fact about the data, and dropping it would conceal that fact.

B. The Synthetic Stand-In, and the Status of Every Number Here

All numbers in this manuscript were computed on a deterministic synthetic dataset generated to resemble published summaries of the real file. This was a project constraint rather than a methodological choice, and we state it plainly: **nothing in Section VI is a finding about maternal health.** The results demonstrate that the pipeline runs, that the metrics behave as designed, and that the analyses produce the shape of answer they claim to.

The generator samples a class from the published class proportions, then samples vitals from class-conditional normal distributions clipped to the published per-column ranges and discretised as a clinic form would record them. It deliberately injects the real file's awkward properties — implausible heart rates, exact duplicate rows, and 7% label noise concentrated on the mid/high boundary — because a stand-in cleaner than the real data would let the pipeline pass tests it would fail in practice. Class-conditional centres were chosen so that guideline thresholds *partially* agree with the labels: perfect agreement would render the calibration study vacuous, and total disagreement would mean measuring the generator rather than anything real.

Synthetic provenance propagates into every artefact — the metadata record, every generated table, the interface banner, and the figures, where the stamp is burned into the image rather than left in the caption, because a figure pasted into a slide deck loses its caption long before it loses its axes. An automated test asserts this propagation.

V. METHOD

A. Guideline Criteria as a First-Class Baseline

We encode eleven obstetric criteria drawn from published guidance, each stored as data with its threshold, severity, and source, so that a clinician can review the rule set without reading code:

- **Hypertension** [21]: systolic 140–159 or diastolic 90–109 mmHg (moderate); systolic $\geq$ 160 or diastolic $\geq$ 110 (severe).
- **Hyperglycaemia in pregnancy** [20]: 8.5–11.0 mmol/L (moderate); ≥ 11.1 (severe), under the OGTT reading of Section IV-A.
- **Fever** [22]: 100.4–101.9°F, i.e. $\geq$ 38°C (moderate); $\geq$ 102°F (severe).
- **Tachycardia**: resting heart rate $>$ 100 bpm (moderate).
- **Maternal age** [23]: $<$ 18 or $\geq$ 35 years (moderate).

These combine through an escalation rule chosen to be defensible rather than clever: any severe feature yields high risk; two or more moderate features yield high risk; exactly one yields mid risk; none yields low risk.

This baseline serves two purposes. As a comparator it sets the floor a learned model must clear to justify its complexity. As an instrument it lets us ask whether the dataset's labels agree with published guidance at all (Section VI-B).

B. Safety-First Metrics

Let the ordinal classes be indexed 0 (low), 1 (mid), 2 (high). We report, in this order:

- **High-risk recall**: the fraction of high-risk mothers escalated at all.

- **Critical-miss rate**: the fraction of high-risk mothers predicted *low* risk,

$$\mathrm{CMR} = \frac{|\{i : y_i = 2 \wedge \hat{y}_i = 0\}|}{|\{i : y_i = 2\}|}. \tag{1}$$

  Reported separately from recall because "mid risk" still escalates her while "low risk" does not — the difference between a delayed referral and no referral.
- **Expected cost**: mean cost under an asymmetric matrix C with $C_{2,0} = 25$, $C_{2,1} = 8$, $C_{1,0} = 4$, and over-escalation priced at 1–2. The values encode a clinical judgement, not a measurement; they are isolated in configuration so they can be challenged.
- **Referral rate**: the share of mothers escalated above low risk. This is the operational cost, and it is a first-class metric because the quiet way a screening tool fails is by referring so many mothers that the programme stops believing it.
- Conventional metrics — accuracy, balanced accuracy, macro F1, and quadratic-weighted κ [24] — reported for comparability with published work, not for selection.

C. A Decision Rule Separate from the Model

Fitting a classifier and deciding where to place the referral threshold are different acts, and conflating them is how safety-critical thresholds end up at whatever `argmax` happened to produce. Given predicted class probabilities $p = (p_0, p_1, p_2)$ and thresholds $(t_{\mathrm{high}}, t_{\mathrm{esc}})$, we assign

$$\hat{y} = \begin{cases} 2 & \text{if } p_2 \geq t_{\mathrm{high}}, \\ 1 & \text{else if } p_1 + p_2 \geq t_{\mathrm{esc}}, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

A 40% chance of high risk therefore escalates even when low risk holds a 45% plurality — a decision `argmax` cannot express.

Thresholds maximise high-risk recall *subject to a referral budget* (55%), ties broken on expected cost. The constraint is part of the objective rather than a footnote: without it the optimum is always "refer everybody", which has perfect recall and is switched off within a month.

D. Protocol, Leakage Control, and the Matched Comparison

We use repeated stratified cross-validation, 5 folds $\times$ 4 repeats. Because the referral thresholds are *fitted parameters*, tuning them on the rows used to score the model would leak. Each outer fold therefore tunes its operating point on inner 3-fold cross-validated probabilities computed from that fold's training portion only, then applies the frozen point to the held-out fold.

The model zoo comprises a majority-class baseline, the guideline baseline, logistic regression, random forest, XGBoost [25], and a small multilayer perceptron, all built on scikit-learn [26], with class balancing throughout. Selection is on mean high-risk recall, tie-broken on expected cost, with baselines excluded from selection but reported alongside.

The guideline rule has no threshold to tune; it refers whoever it refers. Comparing it against a threshold-tuned model at each system's own operating point is therefore not a comparison — whichever refers more people wins on recall by construction. We consequently push every learned model to the guideline baseline's exact referral load and compare there.

E. Attribution That Names Itself

Attributions come from exact SHAP where an exact explainer exists — TreeSHAP for the ensembles, the closed-form linear explainer for logistic regression — and degrade to permutation importance or occlusion otherwise. The method is named in every output and every figure caption. An attribution silently swapped beneath a plot labelled "SHAP" is worse than no plot.

Two implementation traps are worth recording. First, SHAP 0.49 cannot parse the vector-valued `base_score` that XGBoost ≥ 3.0 writes for multi-class models; we route through XGBoost's own `pred_contribs`, which computes the same exact values. Second, a linear SHAP explainer constructed with the explained row as its own background returns *all-zero* attributions — silently, with no error — because every feature equals its own background mean. Both are covered by regression tests.

F. Communication as Part of the System

Model output becomes a bilingual message through hand-authored templates. Urdu is written by hand, not machine-translated, because a mistranslated clinical instruction is a safety incident and because register matters as much as vocabulary. Three commitments are enforced by lint rules executed in continuous integration, so a well-meaning later edit fails the build rather than shipping:

1) **No catastrophising.** Urgency lives in the recommended action and its timeframe ("today", "within a few days"), never in frightening adjectives. A family that panics may go nowhere, or to the wrong place.
2) **No diagnosis.** The tool reports observations and names who to see. It never names a condition, because it cannot confirm one.
3) **Family-inclusive framing at the high band.** Where care decisions are made collectively, a message addressed only to the patient can stall against a household that was never consulted.

A fourth rule emerged from reviewing rendered output rather than from design: **the low-risk band mentions no drivers at all.** Telling a mother her result is normal and then listing which of her vitals "stood out" plants worry the result does not justify, and worry is what makes a family ignore the next message.

VI. Results

All results below are computed on the synthetic stand-in (Section IV-B).

TABLE III
CROSS-VALIDATED PERFORMANCE, 5 FOLDS × 4 REPEATS (MEAN ± STD). ARROWS INDICATE THE DIRECTION OF IMPROVEMENT. BOLD MARKS THE BEST VALUE PER COLUMN.

Model	High-risk recall ↑	Critical miss ↓	Expected cost ↓	Referral rate ↓	Balanced acc. ↑	Accuracy ↑
Guideline baseline	0.900 ± 0.041	0.036 ± 0.027	1.010 ± 0.221	0.633 ± 0.019	0.656 ± 0.031	0.640 ± 0.033
Logistic regression	0.730 ± 0.059	0.044 ± 0.029	1.290 ± 0.201	0.542 ± 0.019	0.759 ± 0.020	0.764 ± 0.018
Multilayer perceptron	0.716 ± 0.054	0.036 ± 0.022	1.269 ± 0.161	0.548 ± 0.025	0.758 ± 0.020	0.764 ± 0.019
XGBoost	0.663 ± 0.065	0.034 ± 0.024	1.395 ± 0.227	0.535 ± 0.023	0.740 ± 0.027	0.750 ± 0.025
Random forest	0.648 ± 0.047	0.029 ± 0.019	1.383 ± 0.186	0.542 ± 0.024	0.744 ± 0.022	0.755 ± 0.021
Majority baseline	0.000	1.000	8.417 ± 0.034	0.000	0.333	0.370 ± 0.001

TABLE IV
DATASET LABELS VERSUS GUIDELINE BANDS

	Guideline band		
Dataset label	low	mid	high
low (370)	267	88	15
mid (350)	90	121	**139**
high (281)	10	18	253

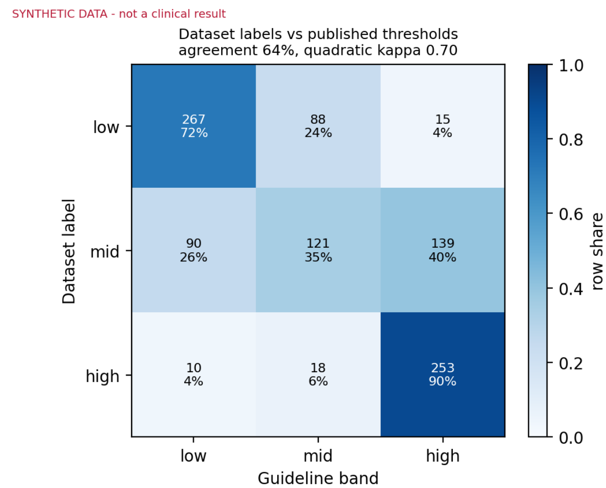

Fig. 1. Dataset labels against guideline bands, normalised by row. Exact agreement 64.0%, quadratic-weighted $\kappa = 0.696$. Computed on synthetic data.

A. Cross-Validated Performance

Table III reports the model zoo under the metric set of Section V-B. Three observations follow.

First, the majority baseline attains 37% accuracy — respectable-looking — while missing every high-risk mother, at roughly six times the expected cost of the worst real model. This is the concrete form of the argument in Section V-B: a metric on which the degenerate model appears merely mediocre rather than disqualified is not a safety metric.

Second, our learned models land in the accuracy band the literature reports (0.750 to 0.764), so this is not a story about a weak implementation. They are simply being asked a different question.

Third, selection on recall changes the answer. Logistic regression is selected — the simplest and most inspectable of the learned models. Note that random forest achieves the *lowest* critical-miss rate (0.029) while having the worst high-risk recall (0.648): it rarely commits the catastrophic error but under-escalates into the mid band more often. Reporting only one of those two numbers would misrepresent it in either direction.

B. Do the Dataset's Labels Agree with Published Guidance?

Exact agreement between guideline bands and dataset labels is 64.0%, with linear $\kappa = 0.584$ and quadratic-weighted $\kappa = 0.696$ — substantial ordinal agreement, but far from the interchangeability that treating these labels as clinical ground truth would assume. Table IV and Fig. 1 give the joint distribution.

The disagreement has structure rather than being symmetric noise.

1) The guidelines are stricter in the middle: Of 350 mothers the dataset labels mid risk, 139 (40%) reach the guideline *high* band. Whatever protocol the labellers used was more permissive than published thresholds on precisely the ambiguous cases.

2) Ten percent of dataset high-risk mothers fall below the guideline high band: Published thresholds alone would not escalate 28 of 281 — a direct argument against replacing a learned model with a rule sheet.

3) A third of the rule set does almost all the work: Table V and Fig. 2 report per-criterion firing rates. The diabetes-in-pregnancy criterion fires for 69.4% of high-risk mothers and 0% of low-risk ones; advanced maternal age for 51.2% versus 4.1%. Two criteria never fire at all, for the reason given in Section IV-A.

4) One undocumented column dominates: Table VI reports the sensitivity of the guideline baseline to the glucose interpretation. Switching from post-load to fasting moves agreement from 64.0% to 31.4%, quadratic κ from 0.696 to 0.136, and the share of mothers escalated from 40.7% to 82.0%.

This is our clearest methodological finding: the sampling condition of a single undocumented column changes the clinical baseline more than any model architecture, hyperparameter, or threshold in the study. Work that trains on this dataset without stating its glucose interpretation has left the most consequential assumption in the analysis unstated.

TABLE V
GUIDELINE CRITERION FIRING RATES (%) BY DATASET LABEL

Criterion	Sev.	All	low	mid	high
Diabetes in pregnancy	S	25.1	0.0	16.0	69.4
Advanced maternal age	M	25.1	4.1	26.3	51.2
Gest. hypertension (dia.)	M	15.5	2.2	9.7	40.2
Gestational diabetes	M	21.1	6.5	36.3	21.4
Gest. hypertension (sys.)	M	10.4	0.0	8.9	26.0
Adolescent pregnancy	M	11.6	14.3	12.3	7.1
Fever	M	4.9	1.9	4.3	9.6
High fever	S	3.0	1.9	1.7	6.0
Severe systolic hypert.	S	1.0	0.0	0.0	3.6
Severe diastolic hypert.	S	0.0	0.0	0.0	0.0
Tachycardia	M	0.0	0.0	0.0	0.0

S: severe; M: moderate. The final two criteria are *silent*: the recorded ranges cannot reach their cut-offs.

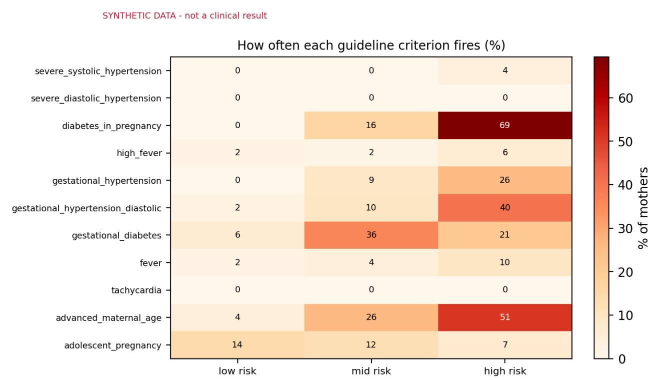

Fig. 2. Per-criterion firing rates by dataset label. Two criteria are unreachable given the recorded ranges and are retained as silent rather than dropped.

TABLE VI
SENSITIVITY TO THE UNDOCUMENTED GLUCOSE SAMPLING CONDITION

Interpretation	Cut-offs	Agreement	κ_q	Flagged high
2-hour OGTT (assumed)	8.5/11.1	0.640	0.696	40.7%
Fasting	5.1/7.0	0.314	0.136	82.0%

C. Does Learning Beat the Guidelines at Equal Cost?

Table VII forces every learned model to the guideline baseline's referral load (≈ 0.648). Fig. 3 shows the full trade-off.

No learned model beats eleven documented thresholds on high-risk recall, even at matched referral load. The gap is not marginal: 0.918 against 0.847 for the best learned model. We report this as the headline because it is the result a reader most needs and the one this literature's framing tends to obscure. On the project's own primary metric, the machine learning did not win.

What the learned models buy is a different error profile. The rule baseline sends 3.5% of high-risk mothers home as low risk; logistic regression sends 1.2%, and both tree ensembles send none. The guideline rule catches more high-risk mothers overall yet is *more* likely to place one in the worst possible band, because a fixed rule has no graded confidence to fall

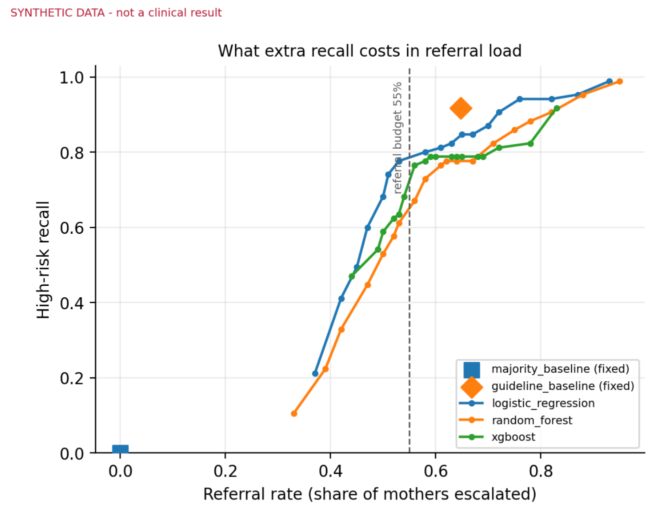

Fig. 3. High-risk recall against referral load. The guideline baseline (marker) sits above every learned model's achievable curve at its own referral rate. The dashed line is the 55% referral budget. Computed on synthetic data.

back on: when no criterion fires, it says low risk with no hedge available. Expected cost is correspondingly lower for logistic regression (0.907) and the MLP (0.887) than for the rule baseline (0.993), and balanced accuracy is substantially better (0.700–0.731 versus 0.642).

The defensible claim is therefore narrower and more interesting than "our model is better": at equal operational cost, the learned models trade a little high-risk recall for near-elimination of the worst individual error and a better overall cost profile. A deployment would sensibly use both — the rules as a non-overridable safety net, the model to grade everything the rules leave silent. *This is the claim most at risk of reversing on real data.*

D. Attribution

Exact LinearSHAP on the selected model ranks blood glucose first by a wide margin, then systolic and diastolic pressure, age, heart rate, and body temperature (Table VIII, Fig. 4). TreeSHAP on the random forest and on XGBoost produces the same ordering. Agreement across three model families and two exact SHAP implementations is a validity signal: glucose and blood pressure dominating a maternal-risk model is clinically expected, and a model leaning on body temperature would have been suspect regardless of its accuracy. The ordering also matches the rule-activity analysis of Section VI-B — two methods, one built from published thresholds and one from fitted coefficients, agreeing on which vitals carry the signal.

E. Threshold Stability

Tuned t_{high} values vary across folds, as expected at $\sim 1{,}000$ rows with roughly 440 rows per inner tuning set. We report the spread rather than an averaged point estimate: a threshold that swings across folds is not one to deploy, and a single mean would hide that.

TABLE VII
PERFORMANCE AT THE GUIDELINE BASELINE'S REFERRAL LOAD

Model	Referral rate	High-risk recall ↑	Critical miss ↓	Expected cost ↓	Balanced acc. ↑
Guideline baseline	0.648	0.918	0.035	0.993	0.642
Logistic regression	0.648	0.847	0.012	0.907	0.700
Multilayer perceptron	0.638	0.835	0.012	0.887	0.731
XGBoost	0.648	0.788	0.000	0.930	0.715
Random forest	0.651	0.776	0.000	0.987	0.699

TABLE VIII
FEATURE ATTRIBUTION, EXACT LINEARSHAP ON THE SELECTED MODEL

Vital	Mean |SHAP|	low	mid	high
Blood glucose	**1.091**	1.637	0.233	1.405
Systolic BP	0.525	0.788	0.152	0.636
Diastolic BP	0.446	0.626	0.044	0.669
Age	0.252	0.379	0.089	0.289
Heart rate	0.190	0.285	0.080	0.205
Body temperature	0.120	0.138	0.042	0.181

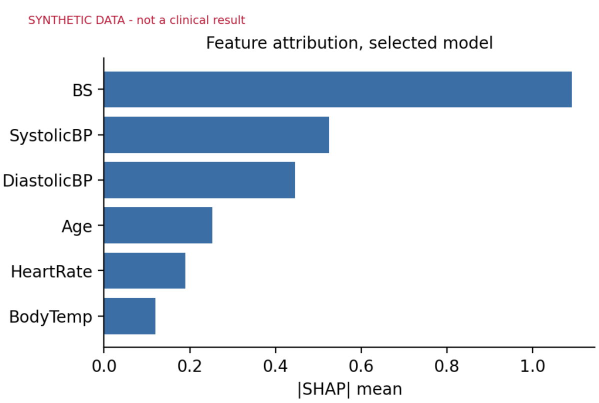

Fig. 4. Feature attribution for the selected model. Method named in the artefact rather than assumed.

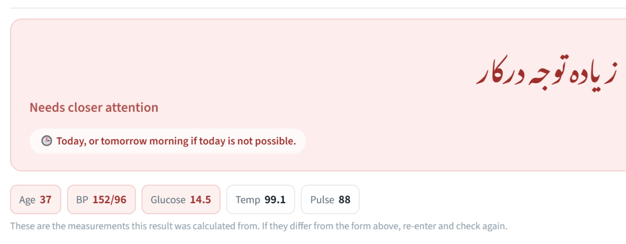

Fig. 5. Rendered high-risk output. The Urdu band name is set in Nastaliq and precedes the English gloss; the timeframe carries the urgency; the measurement chips restate exactly what was scored, with values crossing a guideline threshold highlighted. Interface capture; synthetic data.

F. Message Safety

All three band templates pass the lint gate in both languages. The suite includes adversarial tests that inject a catastrophising phrase, a diagnosis claim, and an action-free message, and assert the linter catches each — the linter is verified to have teeth rather than assumed to. Review status is `UNREVIEWED` for all three bands, and a test asserts the artefact does not claim otherwise. Fig. 5 shows the rendered high-risk output.

VII. DISCUSSION

The negative result is the useful one. A study designed to show that a model beats a rule sheet, and which found the opposite, is more informative than one reporting 84% accuracy and stopping. It also carries a direct design consequence: run the documented criteria as a non-overridable safety net and use the model to grade the cases where no criterion fires. That hybrid is invisible to the accuracy-reporting framing, because that framing never puts the rule baseline on the table.

Undocumented units are a bigger risk than model choice. Section VI-B found the glucose sampling condition moves the clinical baseline far more than anything we varied in modelling. The lesson is unglamorous: for small clinical tabular datasets, effort spent resolving provenance with whoever collected the data will usually beat effort spent on architecture. This extends the datasheet literature [18] with a concrete recipe for the case where documentation is already unrecoverable — state the assumption, then measure what it costs.

Label transfer is measurable, and cheap to measure. Quadratic κ of 0.696 between dataset labels and published thresholds is substantial agreement with a structured residue: the guidelines are systematically stricter on ambiguous mid cases, so a model trained on these labels inherits the labellers' permissiveness. This is an afternoon's work and it is what TRIPOD asks for [13].

Interpretability was not a sacrifice. Logistic regression was selected on the project's own safety criterion, not chosen for simplicity. On six tabular vitals that is the outcome [17] predicts, and it means the deployed model has exactly computable attributions and reviewable coefficients rather than a post-hoc story about a black box.

Communication is not downstream polish. The one design rule we did not anticipate — suppressing driver phrases at the low band — came from reading rendered Urdu, not from the model. It is invisible to every metric in Section VI and would plausibly matter more to real-world outcomes than the difference between our best and worst classifier. Encoding message constraints as CI lint rather than as review guidelines is the component of this work we would most readily reuse elsewhere.

VIII. Threats to Validity

Construct validity. The cost matrix is a judgement, not a measurement. Pricing a critical miss at $25\times$ an unnecessary referral is defensible but arbitrary, and expected-cost comparisons move with it. It is isolated in configuration so it can be challenged; a sensitivity sweep is the obvious next step.

Internal validity. Threshold tuning is nested to prevent leakage, but with $\sim 1{,}000$ rows every reported difference carries wide uncertainty and several gaps in Table III lie within one standard deviation. The matched-referral comparison uses a single held-out split at a fixed operating point, deliberately outside the tuning loop, and is therefore noisier than the cross-validated table.

External validity. All numbers come from synthetic data (Section IV-B). Beyond that, the real dataset is Bangladeshi and nothing here establishes transfer to Pakistan: the calibration study *measures* the gap between labels and published thresholds; it does not close it. Six features are available — no gestational age, parity, obstetric history, proteinuria, or haemoglobin, several of which are more clinically informative than anything in this feature set. The ceiling is set by the data, not the models.

Conclusion validity. The Section VI-C finding is the most likely to reverse on real data, and it is the paper's headline. We flag it as conditional rather than burying the caveat.

Deployment validity. The Urdu templates are `UNREVIEWED` — not signed off by a native-speaker reviewer or a clinician, and never tested with a health worker or a patient.

IX. Conclusion and Future Work

Recast as a decision problem rather than a classification benchmark, the UCI maternal health dataset yields a different and more useful set of answers. Eleven documented obstetric thresholds beat every model we trained on high-risk recall even at matched referral load; the learned models earn their place by nearly eliminating the catastrophic high-to-low error rather than by being more accurate; the dataset's labels agree with published guidance at quadratic $\kappa = 0.696$ with a systematic permissiveness on ambiguous cases; and the undocumented sampling condition of a single glucose column moves the clinical baseline more than any modelling choice available to us.

None of these findings is reachable from an accuracy number, and none required a larger dataset or a better model — only asking what the tool is for, and measuring that instead.

Future work follows directly. The study should be re-run on the real dataset and Section VI-C re-derived. The glucose sampling condition should be resolved with the original collectors, which is the single highest-value action available. The cost matrix should be swept to establish which conclusions are robust to the clinical judgement encoded in it. Validation on Pakistani data is the only way to close rather than measure the transfer gap. The hybrid design suggested by Section VI-C should be formalised and evaluated as a system rather than inferred from two tables. The Urdu should be reviewed by a native speaker and a clinician and tested with LHWs. Finally, accountability for a missed case should be established before the tool reaches anyone's hands: a screening tool without a named owner for its failures is not deployable regardless of its metrics.

Reproducibility

A single seed (20260822) governs data generation, splits, and every model. Each run writes a metadata record containing the data SHA-256, row counts, cross-validation configuration, glucose interpretation, selected model, frozen operating point, attribution method, template review status, and library versions. Every table in Section VI is regenerated from those artefacts by a single script. The test suite comprises 155 tests covering the metric definitions against hand-worked cases, every guideline threshold at its boundary, the message-safety lint gate including adversarial cases, the degenerate-attribution regression, and provenance propagation into artefacts.

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